water, slow interconversion of α-D glucose and β-D glucose via open chain form occurs until equilibrium is established giving a constant specific rotation + 53°. This phenomenon is called **mutarotation**.

Epimers and epimerisation:

Sugar differing in configuration at an asymmetric centre is known as epimers. The process by which one epimer is converted into other is called epimerisation and it requires the enzymes epimerase. Galactose is converted to glucose by this manner in our body.

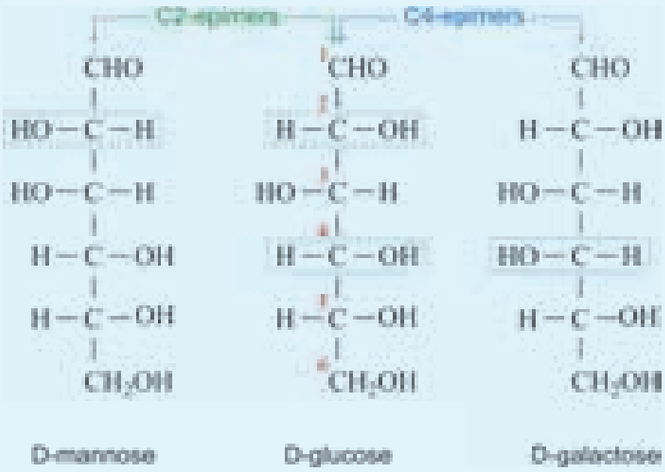


Figure 14.6 Epimers: glucose and mannose are epimers at C2 carbon and glucose and galactose are epimers at C4 carbon

14.1.4 Fructose

Fructose is another commonly known monosaccharide having the same molecular formula as glucose. It is levorotatory and a ketohexose. It is present abundantly in fruits and hence it is also called as fruit sugar.

Preparation

1. From sucrose

Fructose is obtained from sucrose by heating with dilute H_2SO_4 or with the enzyme invertase

$$\underset{\text{Sucrose}}{C_{12}H_{22}O_{11}} + H_2O \xrightarrow[\text{or Invertase}]{H_2SO_4} \underset{\text{Glucose}}{C_6H_{12}O_6} + \underset{\text{Fructose}}{C_6H_{12}O_6}$$

Fructose is separated by crystallisation. The mixture having equal amount of glucose and fructose is termed as invert sugar.

2. From Inulin

Fructose is prepared commercially by hydrolysis of Inulin (a polysaccharide) in an acidic medium.

$$\underset{\text{Inulin}}{(C_6H_{12}O_5)_n} + n\,H_2O \xrightarrow{H^+} n\ \underset{\text{Fructose}}{C_6H_{12}O_6}$$

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